

```
jishnusaurav@jishnusaurav-IdeaPad-L340-15IRH-Gaming:~$ sudo systemctl status mariadb
● mariadb.service - MariaDB 10.1.48 database server
   Loaded: loaded (/lib/systemd/system/mariadb.service; enabled; vendor preset: enabled)
   Active: active (running) since Wed 2021-10-27 00:20:45 IST; 1h 1min ago
     Docs: man:mysqld(8)
           https://mariadb.com/kb/en/library/systemd/
   Process: 1396 ExecStartPost=/bin/sh -c systemctl unset-environment _WSREP_START_POSITION (code=exited, status=0/SUCCESS)
   Process: 1386 ExecStartPost=/etc/mysql/debian-start (code=exited, status=0/SUCCESS)
   Process: 1142 ExecStartPre=/bin/sh -c [ ! -e /usr/bin/galera_recovery ] && VAR= || VAR=/cd /usr/bin/..; /usr/bin/galera_recovery; [ $? -eq 0 ] && systemctl set-environment _WSREP_START_PO
   Process: 1133 ExecStartPre=/bin/sh -c systemctl unset-environment _WSREP_START_POSITION (code=exited, status=0/SUCCESS)
   Process: 1128 ExecStartPre=/usr/bin/install -m 755 -o mysql -g root -d /var/run/mysqld (code=exited, status=0/SUCCESS)
 Main PID: 1242 (mysqld)
   Status: "Taking your SQL requests now..."
     Tasks: 27 (limit: 4915)
   CGroup: /system.slice/mariadb.service
           └─1242 /usr/sbin/mysqld

Oct 27 00:20:44 jishnusaurav-IdeaPad-L340-15IRH-Gaming systemd[1]: Starting MariaDB 10.1.48 database server...
Oct 27 00:20:44 jishnusaurav-IdeaPad-L340-15IRH-Gaming mysqld[1242]: 2021-10-27 0:20:44 139978850233472 [Note] /usr/sbin/mysqld (mysqld 10.1.48-MariaDB-0ubuntu0.18.04.1) starting as process 1242
Oct 27 00:20:45 jishnusaurav-IdeaPad-L340-15IRH-Gaming /etc/mysql/debian-start[1389]: Upgrading MySQL tables if necessary.
Oct 27 00:20:45 jishnusaurav-IdeaPad-L340-15IRH-Gaming systemd[1]: Started MariaDB 10.1.48 database server.
Oct 27 00:20:45 jishnusaurav-IdeaPad-L340-15IRH-Gaming /etc/mysql/debian-start[1393]: /usr/bin/mysqld upgrades: the '--basedir' option is always ignored
Oct 27 00:20:45 jishnusaurav-IdeaPad-L340-15IRH-Gaming /etc/mysql/debian-start[1393]: Looking for 'mysqld' as: /usr/bin/mysqld
Oct 27 00:20:45 jishnusaurav-IdeaPad-L340-15IRH-Gaming /etc/mysql/debian-start[1393]: Looking for 'mysqldcheck' as: /usr/bin/mysqldcheck
Oct 27 00:20:45 jishnusaurav-IdeaPad-L340-15IRH-Gaming /etc/mysql/debian-start[1393]: This installation of MySQL is already upgraded to 10.1.48-MariaDB, use --force if you still need to run mysq
Oct 27 00:20:45 jishnusaurav-IdeaPad-L340-15IRH-Gaming /etc/mysql/debian-start[1415]: checking for insecure root accounts.
[lines 1-25/25 (END)]
```

Figure 6: Checking the status of MariaDB

```
jishnusaurav@jishnusaurav-IdeaPad-L340-15IRH-Gaming:~$ sudo mysql_secure_installation
[sudo] password for jishnusaurav:

NOTE: RUNNING ALL PARTS OF THIS SCRIPT IS RECOMMENDED FOR ALL MariaDB
SERVERS IN PRODUCTION USE! PLEASE READ EACH STEP CAREFULLY!

In order to log into MariaDB to secure it, we'll need the current
password for the root user. If you've just installed MariaDB, and
you haven't set the root password yet, the password will be blank,
so you should just press enter here.

Enter current password for root (enter for none):
OK, successfully used password, moving on...

Setting the root password ensures that nobody can log into the MariaDB
root user without the proper authorisation.

You already have a root password set, so you can safely answer 'n'.

Change the root password? [Y/n] y
New password:
Re-enter new password:
Password updated successfully!
```

Figure 7: Securing the database

```
jishnusaurav@jishnusaurav-IdeaPad-L340-15IRH-Gaming:~$ sudo mysql
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 34
Server version: 10.1.48-MariaDB-0ubuntu0.18.04.1 Ubuntu 18.04

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]>
MariaDB [(none)]>
MariaDB [(none)]>
MariaDB [(none)]>
MariaDB [(none)]>
MariaDB [(none)]>
```

Figure 8: Starting the DB on command line

```
MariaDB [(none)]> show databases;
+-----+
| Database |
+-----+
| information_schema |
| mysql |
| performance_schema |
+-----+
3 rows in set (0.00 sec)
```

Figure 9: Checking the existing databases

First run the command, 'sudo mysql', as shown in Figure 8.

We can now check the existing databases (as shown in Figure 9), using the following command:

```
show databases;
```

Now, to create a database, we can

```
MariaDB [(none)]> create database jishnu
-> ;
Query OK, 1 row affected (0.00 sec)

MariaDB [(none)]> show databases
-> ;
+-----+
| Database |
+-----+
| information_schema |
| jishnu |
| mysql |
| performance_schema |
+-----+
4 rows in set (0.01 sec)
```

Figure 10: Creating a database and checking the newly created one

```
MariaDB [(none)]>
MariaDB [(none)]> exit
Bye
```

Figure 11: Exiting MariaDB from the command line

use the following command just like MySQL, as shown in Figure 10:

Create database jishnu;

Now we can use our database normally. To exit the command line, we can just use the `exit` command (as shown in Figure 11).

We have now understood how to install, use and secure our MariaDB database. It can be great fun to use new database servers and explore them. Do give this one a try! **END** 🐼

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The author is a full stack Web developer, interested in modern technologies like artificial intelligence and machine learning. He is currently working in the field of natural language processing.